

Extended Data 9 | Per-dataset summary statistics

dataset	cells	packing median (cells/mm ²)	packing p90	median NN (um)	median tx/cell
Xenium breast	167,780*	6,534	11,956	9.43	164
Xenium lung	150,365*	7,285	n/a	n/a	132
Xenium colorectal	388,175†	17,424	33,607	5.58	78
MERFISH hypothal.	100-108/FOV‡	2,525	n/a	11.96	280

* cell count from committed report (GATE2/GATE3_REPORT). † colorectal count: 10x portal (dataset provenance, not a committed CSV). ‡ MERFISH: committed per-FOV counts (realism.csv); section total not in a committed file. Lung median NN was computed but not committed to CSV (shown n/a). Packing/NN/tx are committed-CSV values (lever_ceiling.csv, gate3_pin_lung.csv, realism.csv).